B.VOC. (HOSPITAL ADMINISTRATION AND MANAGEMENT)

Title of the Programme: **Bachelor of Vocation in Hospital Administration and Management leading to degree "B.Voc. (Hospital Administration and Management)**

We are today living in a world that has its fair share of challenges. As life is competitive, stressful and demanding and the adulterated environment hardly proves of any help, lifestyle diseases are a rampage. Also adulteration, pollution and increasing sustainability issues have constantly given birth to new forms of life threatening diseases. No wonder hospitals have been establishments that have constantly been on an increase with the demand of facility increasing each passing day.

Medical establishments such as hospitals, clinics, rehabilitation centres and so on are looked upon as edifices of hope by the diseased and the ill. Like any other business, medical establishments are organized institutions. They follow complex processes and need efficient manpower to manage these processes. Organization becomes they key in such institutions because disorganization would spell disaster. It would inevitably mean that so many lives are being constantly put at stake. And each life is precious national property.

Hospital Administration and management becomes a very different field of organization in the sense that it is technical and needs at least a considerable knowledge of the medical intricacies. It is this demand that candidates trained in Hospital management and administration are fully equipped to deal with. Hospital Administration and management as a course is concerned with the organization, coordination, planning, staffing, evaluating and controlling of health services for the masses. The primary objective is to provide quality healthcare to people. Professional hospital administrators have proven how institutions can be managed proficiently, economically and successfully.

There has been seen a remarkable growth in the hospital industry in India, which has lead to a great demand and popularity of the hospital management related courses. Earlier senior doctors were responsible to perform the role of a hospital manager apart from their normal medical duties. However, times have changed. Earlier there were just hospitals. Now, hospitals themselves have been categorized as ordinary hospitals, specialty and super specialty hospitals.

The requirement of professional administrators in the hospitals is growing briskly mostly because of the already mentioned reason that the nature of work in hospitals is quite different from other organizations. Hospitals are expected to deliver quality service 24 x 7 x 365, essentially because lives are at stake.

The urgent nature of its work and the level of efficiency that is expected have increased the need of well-formulated hospital management courses. What's more, a career in hospital

management helps you to achieve two goals in life, namely: a good pay package and service to mankind.

Moreover, there is manifold increase in the complexity and processes that take place in the successful management of a modern day hospital. This in turn has lead to a tremendous demand for efficient professionals who can handle the day-to-day chores of hospital management and administration. Highly trained professionals are needed for the smooth operating of a hospital. To be a successful professional in this field you should have a service-oriented bent of mind and the capacity to work for long hours. Furthermore, you must have a high emotional quotient to survive the highly charged atmosphere of a hospital. Such attitudes and habit cannot be built in a day. they have to be carefully nurtured and administered over a period of time.

For a career in hospital administration, you should possess some unique qualities such as an innovative attitude, responsibility, self motivation and selfless devotion towards humanity.

As health services and medical managers are responsible for hundreds of employees and very expensive equipment and facilities, you should be able to make effective decisions, interpret data and understand information systems and finance. Since you will be dealing with people all the time you must have strong communication skills.

Other essential traits include skill to take swift decisions, a friendly approach towards work and the ability to handle work pressure and the public at large. The ability to cope up with deadlines, have excellent communication skills and leadership qualities is also a very important aspect. Other essential traits include skill to take swift decisions, a friendly approach towards work and the ability to handle work pressure and the public at large. The ability to cope up with deadlines, have excellent communication skills and leadership qualities is also a very important aspect.

The healthcare concept in India has undergone a tremendous change in recent years. People have become increasingly aware of the importance of healthcare and this has led to higher expectations and an increased demand for a high quality of medical care and facilities.

The importance of health care can never shrink and the number of institutes providing health care is only increasing. There are more than two and half lakhs health care institutes that require quality hospital administrators and managers in India.

The growing need of high professionalism would only increase the importance of hospital management courses in India. Along with the government, various private hospitals are today competing with each other to provide first class health care services to the masses across the country. In this respect, the demand of professional hospital administrators is rising tremendously. The government of India is making special and concerted efforts in providing health care in both rural and urban areas of the country. With more and more private players also entering the hospital industry the demand for efficient administrators and managers is bound to go higher and higher.

There is a massive growth especially in super specialty hospitals due to which the demand for hospital managers and administrators have immensely increased. Moreover a professional

course in hospital management not only opens door for various big and small hospitals but also in corporate, private and public sectors. A large number of private hospitals and clinics have come up all over the country. With increasing emphasis on quality of health care and patient satisfaction, there is a remarkable need for persons with a professional qualification in hospital management.

Not only this, the sector also offers prospects of global recognition with medical tourism becoming more widespread. If you're talented and demonstrate a good work ethic, you can advance quickly. There is no limit to where your dreams and ambitions can take you in the exciting and fascinating world of hospital management and administration.

The salary of an administrator working for a medium grade hospital is around Rs.30,000 per month. The annual remuneration of an administrator in a super specialty 1000-bed hospital can easily go into lakhs of rupees.

Thus to sum up there is no dearth of employment opportunities in employment for those candidates opting for the course. The Hospital Manager's role differs according to the size of the organization. In a smaller, medical or health care center, they have to handle day-to-day administrative work, which may involve billing, maintenance, equipment supply etc. and also work closely with doctors, in group practice and decision-making. But large-scale organizations may require the services of a manager in key aspects as well. Hospital administrators have to co-ordinate amongst the various departments of the hospital and have to ensure that all equipments are functional. They are responsible for hiring non-medical staff, maintenance of hospital services, security, building the hospital brand, non-marketing, accounts, financial operations, planning and control for continuous supply of materials, management of the research and clinical work etc.

Fresh degree holders may start their professional career as assistant hospital administrators or as managers of non-health departments like finance. One can find employment in large corporate and public sector hospitals, international and national healthcare organizations and health insurance companies, College and Universities, Healthcare Portals, Nursing Homes, Mental Health Facilities, Pharmaceuticals and Hospital Supply Firms, Medical Software Companies and Hospital Consulting Firms. Big hospitals like Escorts Heart Institute, Apollo Hospital, Belleview Hospital, CMRI, AMRI, Moolchand Hospital, Rajiv Gandhi Cancer Hospital etc. appoint hospital management graduates for their administration purposes.

B.Voc. (HOSPITAL ADMINISTRATION AND MANAGMENT)

Semester I								
Paper Code	**Title**	**Generic/Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credits**
*GEN - 101	Communication Skills	Generic Component	Theory	20	80	----	----	6
*GEN-102	Fundamentals of Information Technology	Generic Component	Theory	20	80	----	----	6
HAM - 103	Hospital Administration	Skill Component	Theory & Practical	10	40	10	40	6
HAM - 104	Supportive Services and Facilities Management	Skill Component	Theory & Practical	10	40	10	40	6
HAM - 105	Hospital & Health Care System	Skill Component	Theory & Practical	10	40	10	40	6
HAM-106	Hospital Training & Viva-Voce	--	--	--	--	--	--	Non Credit

Semester II

Paper Code	Title	Generic/Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credits
*GEN - 201	Soft Skills and Personality Development	Generic Component	Theory	20	80	----	----	6
HAM-108	Accounting for Decision Making	Generic Component	Theory	20	80	----	----	6
HAM-109	Patient Care Planning & Management	Skill Component	Theory & Practical	10	40	10	40	6
HAM 110	Clinical, Diagnostic & Therapeutic Services	Skill Component	Theory & Practical	10	40	10	40	6
HAM-111	Managing Quality in Health Care	Skill Component	Theory & Practical	10	40	10	40	6
HAM - 112	Hospital Training & Viva Voce	--	--	--	--	--	--	Non Credit

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

Semester III

Paper Code	Title	Generic/Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credits
*GEN 301	Value Education And Human Rights	Generic Component	Theory	20	80	----	----	6
HAM 202	Human Resource Management & Drug Management	Generic Component	Theory	20	80	----	----	6
HAM 203	Nursing Administration and Hospital Management	Skill Component	Theory & Practical	10	40	10	40	6
HAM204	Patient Care & Behaviour	Skill Component	Theory & Practical	10	40	10	40	6
HAM205	Management of Information System in Hospital	Skill Component	Theory & Practical	10	40	10	40	6
HAM206	Hospital Training & Viva-Voce			--	--	--	--	Non Credit

Semester IV

Paper Code	Title	Generic/Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credits
*GEN 401	Environmental Studies	Generic Component	Theory	20	80	----	----	6
HAM208	Public Relations	Generic Component	Theory	20	80	----	----	6
HAM209	Management of Out Patient Department (OPD)	Skill Component	Theory & Practical	10	40	10	40	6
HAM 210	Healthcare Marketing	Skill Component	Theory & Practical	10	40	10	40	6
HAM 211	Medical Equipment & Floor Management	Skill Component	Theory & Practical	10	40	10	40	6
HAM212	Hospital Training & Viva-Voce	--	--	--	--	--	--	Non Credit

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

Semester V

Paper Code	Title	Generic/Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credits
*GEN 501	Critical Thinking And Elementary Statistics	Generic Component	Theory	20	80	----	----	6
HAM 302	Medical Ethics & Legal Issues in Hospital	Generic Component	Theory	20	80	----	----	6
HAM 303	Hospital Waste Management	Skill Component	Theory & Practical	10	40	10	40	6
HAM 304	Hospital Safety System and Health Care Standards	Skill Component	Theory & Practical	10	40	10	40	6
HAM 305	Hospital Material Management	Skill Component	Theory & Practical	10	40	10	40	6
HAM 306	Hospital Training & Viva-Voce	--	--	--	--	--	--	Non Credit

Semester VI

Paper Code	Title	Generic/Skill Component	Theory/ Practical	Internal (Theory)	External (Theory)	Internal (Practical)	External (Practical)	Credits
*GEN 601	Entrepreneurship Development Programme	Generic Component	Theory	20	80	----	----	6
HAM308	Global Healthcare Scenario & Insurance	Generic Component	Theory	20	80	----	----	6
HAM309	Hospital Planning & Management	Skill Component	Theory & Practical	10	40	10	40	6
HAM310	Computers in Medicine	Skill Component	Theory & Practical	10	40	10	40	6
HAM 311	Research Methodology	Skill Component	Theory & Practical	10	40	10	40	6
HAM 312	Hospital Training & Viva-Voce	--	--	--	--	--	--	Non Credit

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

At the end of the semester the students should compulsorily undergo two week Hospital training HAM 106, HAM 112, HAM 206, HAM 212, HAM 306 & HAM 312) and the relevant certificate has to be submitted

HAM-103-Hospital Administration

Objective:

This course intends to familiarize and develop understanding of the students to general administration of the hospital designing for optimum utilization of hospital resources and uninterrupted delivery of healthcare services.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-
Office Assistant, Receptionist, Data Entry Operator, Ward Assistant, Supportive Services, Assistant, Assistant Quality Manager.

Unit – I

1. Routine Admission/Discharge Procedures/Discharge Summary
2. Hospital Utilization Statistics - Average Length of Stay (ALS), Bed Occupancy Rare, Turn Over Interval
3. Daily Reports / Returns - Hospital Census, Matron's Report, Medical Officer's Report, Casualty Report, Medico-Legal Cases, Report from ICU / ICCU, Security Report, Maintenance Department Report OT List
4. Patient's Complaints
5. Medical Certificates

Unit II

1. **Hospital Committees -** Role, Composition, Frequency of Meetings, Minutes of the Meetings, Follow up Actions.
2. **Patient Satisfaction Survey -** Interviews, Questionnaires, Observations, Group Discussions, Patient Opinion Polls, Report Writing.
3. **Duty Roster of various categories of Staff.**
4. **Availability of Materials -** Critical Items, Stock Level, Procurement Methods.

Unit – III

1. Administration of Patient Related Schemes - Medical Insurance (Cashless Benefit), CGHS, ECHS, CSMA, TPA, ESI.
2. Front Office : Duties & Responsibilities
3. Duties & Responsibilities of the Hospital Administrator/CEO - In Profit Making Hospitals, In Non-Profit Making Hospitals
4. Disaster Management/Disaster Plan
5. Marketing of Hospital

Unit IV

1. **Hospital Security -** Staff, Patients, New born babies, Female staff/Patients, Stores.
2. **Application of Hospital Information System (HIS) & Management Information System (MIS)**
3. **Negotiation Skills -** Purchase of Stores / Equipment, Union Matters, Collective Bargaining.

References

- Stephen P. Robbins and Mary Coulter, **Management** (Prentice Hall of India Pvt. Ltd., New Delhi)
- J.E. Park and K. Park, **Textbook of Preventive and Social Medicine** (M/S Banarsidas Bhanot Publishers, Jabalpur)
- Elaine La Monica, Management in Health Care (Macmillan Press Ltd, London)
- B.M. Sakharkar, **Principles of Hospital Administration and Planning** (Jaypee Brothers Medical Publishers Pvt. Ltd., New Delhi)

HAM-104-SUPPORTIVE SERVICES AND FACILITIES MANAGEMENT

Objective:

The objective of the course is to equip the learners with decision making skills in planning, scheduling and control of production and operation functions in hospitals.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-
Office Assistant, Receptionist, Data Entry Operator, Ward Assistant, Supportive Services, Assistant, Assistant Quality Manager.

UNIT –I

Front office – Admission – Billing – Medical Records – Ambulatory Care – Death in Hospital – Brought-in Dead. Maintenance and Repairs Bio Medical Equipment. Clinical Services – Clinical Departments – Out patient Department (OPD) – Introduction – Location – Types of patients of OPD – Facilities – Flow Pattern of patients – Training and Coordination;

Unit II

Radiology – Location – Layout – X-ray Rooms – Types of X-ray machines – Staff – USG – CT – MRI – ECG, PET Scan
Supporting Services – House Keeping – Linen and Laundry – Food Services – Central Sterile Supply Department (CSSD), Maintenance

UNIT –III

To consider various operational aspects of important services - Front office and medical record services- Patient Admission / Discharge -All patients related services and assistance. Good communication.-Nursing care with full devotion / commitment-Diagnostic Services - Radiologist, Pathologists and Lab Technicians-Sterilization Services -Blood transfusion services-Proper and respectful disposal of deceased person

Unit IV

Nutrition and Dietary services - Pharmacy services - Medical records services-Laundry Services - House keeping services - CSSD energy conservation methods – AMC-Ambulance Services - Mortuary services - Hospital security services-Disaster Management - Fire Hazards - Engineering Hazards - Radiological Hazards - Outsourcing of Support services - few case studies.

References:

- Maduri Sarma, Essentials for Hospital Support Services and Physical Infrastructure, Jaypee Brothers Publications.
- Prabhu KM, Sood SK, Hospital Laboratory Services Organization and Management, Journal of Academy of Hospital Administration.
- Srinivasan, A.V. (ed.), Managing a Modern Hospital, Chapter 12, Response Books, New Delhi.
- Schoedev, Roger G., Operations Management – Decision Making in Operations Function, McGraw Hill, New Delhi.

HAM-105-HOSPITAL AND HEALTH CARE SYSTEM

Objective:

To make the student to understand health care sector and hospital set up in a larger perspective with an emphasis on the systems.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Office Assistant, Receptionist, Data Entry Operator, Ward Assistant, Supportive Services, Assistant, Assistant Quality Manager.

UNIT I

Health - Definition and Meaning of Health, Basic information relating to health, Definition and meaning of hospital, historical development of hospital, grown of hospital in India, Changing concept of hospital during 1900-2003,

UNIT II

Health The modern hospital, A complex entity, Present status of hospitals in India, Present State of Govt. hospital, National Healthy policy, Role of Hospitals, benefits to the health care system, Overview of Health care delivery system.

Unit III

INTRODUCTION TO HEALTH CARE - A Conceptual Approach to Understanding the Health Care Systems: Evolution – Institutional Settings - Out Patient services – Medical Services – Surgical Services – Operating department – Pediatric services – Dental services – Psychiatric services – Casualty & Emergency services – Hospital Laboratory services – Anesthesia services – Obstetrics and Gynecology services – Neuro – Surgery service – Neurology services.

UNIT IV

OVERVIEW OF HEALTH CARE SECTOR IN INDIA - Primary care – Secondary care – Tertiary care – Rural Medical care – urban medical care – curative care – Preventive care – General & special Hospitals-Understanding the Hospital Management – Role of Medical, Nursing Staff, Paramedical and Supporting Staff - Health Policy - Population Policy - Drug Policy – Medical Education Policy

References:

- Zweifel.Peter, Breyer Friedrich, and MathiasKifmann, 'Health economics', Springer.
- Y.Shanmugasundaram, 'Theory and practice of health economics in India', Institute for Advanced Studies & Research.

Paper - HAM 106

Hospital Training and Viva Voce

At the end of the semester, students should compulsorily undergo two week Hospital training (HAM 106) and the relevant certificate and project report has to be submitted.
Practical training includes the following:

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Office Assistant, Receptionist, Data Entry Operator, Ward Assistant, Supportive Services, Assistant, Assistant Quality Manager.

Reception - Receiving patients and providing information, Providing information of availability of doctors, Fixing appointment with doctors, Arrangement of direct admission for urgent cases, Public Address System, Supervising internal transportation system, No. of Receptionist required to handle the Desk

Registration - Infrastructure – Lay out, Physical facilities, Patient registration, Assisting in collection of hospital charges, Transmitting information to respective doctor, Assisting in sending patient to the respective departments, Computer based functional activity by Receptionist

Admission - Documentation of registration for Admission of patients, Preparation of files with relevant papers, Observing the process of receiving advance payment during admission either by cash or credit card, Formalities of procedures for ensuring availability of payee's fund, Distribution of visiting card, literatures, brochure etc. to the client, Transfer of Patient to ward, Ambulance management and billing, Deployment of no. of Receptionist-cum-Office Assistant, Foreign Exchange facilities

House Keeping - All departments, wards and adjacent areas within the hospital premises, Cleaning process – Moping, Sweeping, Washing, Shampooing (Carpet), Brooming, Process of selection of detergents and disinfectants, Disposal of waste materials, Sources of waste in different areas of hospital, Categories of waste identification in the hospital, Waste segregation according to Biomedical Waste Management & Handling Rules, Application of colour code, including poly pack, bin etc., Waste transportation process, Internal waste storage system, Disposal process of biomedical waste (solid & liquid), Management of infected healthcare worker, Use of Personnel Protective Equipment (PPE), Supervision / leadership style, Staff training for precaution taken, Pest control system, Record maintenance, Waste Tracking, Equipment for wet and dry scrubbing machine, Usage of vacuum cleaner, Garbage Trolley, Roll of - (a) Operation Manager (b) Housekeeping Manager (c) Supervisor (round the clock) in different shift (d) Sweeper (e) Ward boy

SEMESTER II

HAM-109-Patient Care Planning and Management

Course Objectives:

The overall purpose of this course is to enhance the knowledge, skills and attitude in managing clinical services in hospitals. To fulfill this, the objectives of the course are: To understand the role of administrator in patient care planning and management To develop the skills of planning are location, design and layout of clinical services from staff / customer point of view. To develop the skills to optimally utilize the available resources (staff, space, equipment, facilities, supplies, etc.) To understand the need and importance of equipment and physical facilities To understand as well as develop the skills of developing Policy and Procedures for Clinical Services.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Office Assistant, Receptionist, Data Entry Operator, Ward Assistant, Supportive Services, Assistant, Assistant Quality Manager.

Unit I

Hospital Planning - Concept of Planning, Guiding Principles in Planning Hospital Facilities and Services,
Regional Planning and Factors to be emphasized, Steps in Hospital Planning; Planning Team and Stages of Project, Estimation, Architect Brief and Master Plan, Selection of Site and Decision on Land, Space and Utilities

UNIT II

Outpatient Services - Objectives, Functions, Location, Design and Layout, Policy and Procedures, Organization, Staffing, Equipment and Facilities, Key Result Areas and Performance / Quality Indicators
Daily Planning and Scheduling of Work, Managing Time: Waiting Time and Total Time, Specialty, Sub-specialty and Super Specialty Clinics, Diagnosis, Physiotherapy and Occupational Therapy, Emerging Concepts: Day Care, Reservation, Appointment by Phone, Medico-social Works / Patient Counseling
Other Facilities: Pharmacy, Gifts Shop, Prayer / Meditation Room

Unit III

Trauma Care: Emergency and Casualty Services - Objectives, Functions, Location, Design and Layout Policy and Procedures, Organization, Staffing, Equipment and Facilities, Key Result Areas and Performance / Quality Indicators, Disaster Management: Principles and Classification, Life Saving Drugs, Ambulance and Paramedic Services, Medico-legal Procedures, Forms and Registers to be maintained, Communication System
Operation Theatre Objectives Functions Location, Design and Layout Policy and Procedures Organization Staffing Equipment and Facilities Key Result Areas Daily Planning and Scheduling Determinants of number of Operating Rooms Zoning and Aseptic / Sterile Techniques Clinical Protocols Sub-stores, CSSD, Immediate Postoperative Recovery Rooms Safety Issues

UNIT IV

Inpatient Services - Inpatient Care, Objectives, Functions, Locations, Design and Layout, Policy and Procedures, Organization, Staffing, Equipment and Facilities, Key Result Areas and Performance / Quality Indicators, Admission, Transfer, Billing and Discharge Procedures, Managing Deaths, Intensive Care Units, Objectives, Functions, Location, Design and Layout, Policy and Procedures, Organization Staffing, Equipment and Facilities, Keya Result Areas and Performance / Quality Indicators, Types of ICUs

References:

- NHS, Guide to Good Practices in Hospital Administration (Department of Health and Social Security: National Health Services, London)
- R. Liewelyn Davies and HMC Macaulay, Hospital Planning and Administration (Jaypee Brothers Medical Publishers P. Ltd., New Delhi)
- Syed Amin Tabish, Hospital and Health Services Administration Principles and Practice (Oxford University Press, New Delhi)
- B.M. Sakharkar, Principles of Hospital Administration and Planning (Jaypee Brothers Medical Publishers Pvt. Ltd., New Delhi)

HAM-110-CLINICAL, DIAGNOSTIC & THERAPEUTIC SERVICES

Objective:

The Objective of the Course is to expose the learner to the basic concepts of clinical, diagnostic and Therapeutic services for an efficient and effective Administration of Hospitals.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-
Office Assistant, Receptionist, Data Entry Operator, Ward Assistant, Supportive Services, Assistant, Assistant Quality Manager.

UNIT-I

Common Terms of Healthcare Management: Terms related to levels of healthcare; Primary; Secondary and Tertiary – Systems of Medicine – Ayurveda, Siddha, Unani, Homeopathy, Yoga, Naturopathy, Reiki, etc. – Preventive and Curative Medical Care – General and Specialty Hospitals – Pharmacopeia (inpatient and outpatient)

UNIT-II

Specialty-wise terminology – Pathology terms of Common Use – Clinical, diagnostic and therapeutic terms – Pediatric services – Dental – Psychiatric. Casualty and Emergency – Neurology – Obstetrics and Gynecology – Dermatology intensive care – Coronary care services.

UNIT-III

Introduction to medical terminology, Word formation & syntax - Greek alphabet, Greek & Latin prepositional & adverbial prefixes, Singular & plural endings, Commonly used prefixes, suffixes and root words in medical terminology, Common Latin term used in prescription writing, Study of standard abbreviations, Commonly used medical terms to define different parts of the body

UNIT-IV

Medical Records – Admission – Billing – Nursing Records – Diagnostic Records – Infection, Control Records – Maintenance of Intensive Care Units Records. Housekeeping Records – Food Records – Engineering Records – Maintenance Records –Security Records – Fatal Documents – Mortuary Maintenance Records – Transportation – Medico Legal Records.

References:

- Rajiv Mishra, Rachel Chaterjee, Sujatha Rao, India Health Report, Oxford University Press, New Delhi.

HAM-111-MANAGING QUALITY IN HEALTHCARE

Objective:

This course intends to familiarize and develop understanding of the students of the basic principles and practices of quality in health care for efficient administration of hospital and delivery of healthcare services.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Office Assistant, Receptionist, Data Entry Operator, Ward Assistant, Supportive Services, Assistant, Assistant Quality Manager.

Unit I

Fundamentals - History, Need and Importance of Quality Management-Core Values, Concepts and Model-Quality Gurus and their views-Dimensions of Quality- Principles of Quality Management: Structure, Process and Outcome- Quality Vs Productivity Vs Profitability- Cost of conformance and non-conformance to Quality.

UNIT-II

Components and Quality Management Process - Major Components in Quality- Setting Objectives- Quality Investment- Activity Monitoring- Performance Assessment-Quality Management Process- Leadership Commitment- Corporate Framework on Quality-Transformation of Corporate Culture- Customer Focus- Process Focus-Collaborative Approach-Education, Training and Development- Learning by Practice and Teaching- Benchmarking-Quality Measurement and Statistical Report at all Levels- Recognition and Reward-Management Integration.

Unit III

Healthcare Quality - Quality / Customer Service-Define Customer and Identify Customers-Customer Experience:-Core Service & Delivery of Service-Excellent Customer Service; Caring Service- Individual Behaviour: Stress, Communication and Interpersonal Relationship-Patient Satisfaction-Rights and Responsibilities of Patients- Satisfaction and Delight- Quality Indicators of Patient Satisfaction-Clinical Quality-Complication and Infection Rate-.2 Admission, Follow Up and Continuity of Care-Measuring Quality- Setting Objectives and Agreeing upon Standards-Develop Key Result Areas and Performance Indicators- Feedback: Customers, Staff, Suppliers, etc.-Quality Audit and Review Techniques.

UNIT-IV

Organization and Roles in Quality - Quality Policy: Commitment to Patients and Staff-Code of Conduct for Health Professionals- Job Description of Quality Manager- Quality Steering Committee-Quality Council- Quality Teams: Task Force, Quality Circle-Obstacles to Practice Quality.

Recent Trends - ISO Certification- Accreditation: NABH and JCI- Quality Awards Scheme.

References:

- Wilson CRM, Hospital Wide Quality Assurance (Saunders, Ontario)
- Hugh C. H. Kogh, Total Quality Management in Health Care (Longman Publication) ISBN 0582 04696
- Roger Ellis and Dorothy Whittington, Quality Assurance in Health Care – A Hand Book (Edward Ainoid, London) ISBN 0-340-55273-5
- Edward W. Deming, Out of the Crisis (Cambridge University Press, Cambridge)
- Richard Smith (ed.), Audit in Action (British Medical Journal, London) ISBN 0-7279-0317-9
- Nankemp and Eileen Richardson, Quality Assurance In Nursing Practice (Butterworth Heinemann Ltd., London, Second Edition) ISBN 0-7506-2326-8
- Helga Drummond, The TQM Movement – What Total Quality Management is Really All About (UBSPC, New Delhi)

Paper - HAM 112

Hospital Training and Viva Voce

At the end of the semester, students should compulsorily undergo two week Hospital training (HAM 112) and the relevant certificate and project report has to be submitted.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-
Office Assistant, Receptionist, Data Entry Operator, Ward Assistant, Supportive Services, Assistant, Assistant Quality Manager.

Practical training includes the following:
Billing & Discharge Involvement in the process of:- 1) Assimilation of information through LAN from the different departments 2) Checking of data in the computer 3) Observation of different billing system like; a) Cash Billing – As per hospital's prevailing rate schedule b) TPA Billing – As per hospital's prevailing rate schedule c) Copt. Billing – Tailor-made billing as per agreement 4) Handing over the bill to the patient party 5) Providing Birth Certificate, Amputed Certificate, Death Certificate, Referral Letter, as and when required 6) Providing discharge certificate in original along with all relevant documents for cash payee patient and photo copy of discharge certificate to the corporate and TPA patients 7) Deployment of no. of Receptionist-cum-Office Assistant 8) Facilities – Computer, Swapping Machine, Note Counting Machine, Communication facility
Emergency Observation of:- 1) Location of Emergency in hospital 2) Basic lay out of Emergency (a) Receiving patients (b) Patient examination zone (c) Patient investigation zone (d) Procedure room (d) Cardio Pulmonary Resuscitation Unit (e) Dead patient barrier (f) EMO's Office (g) Nursing Station 3) Infrastructural facilities a) Office of EMO b) Nursing Office 4) Equipments 5) Legal procedure system 6) Emergency Operation Theatre (a) Arrangement of instruments and oxygen for minor Operation (b) Sterilization and fumigation of O.T. Room 7) Admission procedure 8) Billing for day care procedure in Emergency Dept. 8) Staffing i) Medical ii) Para Medical 9) Security Critical Care Unit Observation / involvement in:- Physical structure and facilities 1) Organizational structure 2) Bed orientation pattern 3) Lighting system 4) Monitoring system including connection to centralized monitoring system (a) Invasive (i) Central Venus Pressure (CVP) (ii) Arterial Blood Pressure monitoring (iii) Cardiac output monitoring (iv) Pulmonary Arterial monitoring (b) Non-invasive (i) Heart bit rate (ii) Blood Pressure (iii) Respiratory rate (iv) Intake & output monitoring
Maintenance of sterility and general cleanliness 1) Maintenance of sterility of the ward 2) Maintenance of sterility of clothing used in CCU 3) General sterilizing procedures before examination of the patient Administration of CCU 1) Qualified doctor having experience in CCU (as per WHO ratio) 2) Qualified nurse having experience in CCU (as per WHO ratio) 3) Deployment of qualified paramedical staff as per WHO ratio) Workflow 1) In flow of patient 2) Type of patients commonly admitted in CCU 3) Pattern of treatment patient receive in CCU 4) Type of patient actually needs life saving support 5) Out flow of patient 6) Counseling with outside visitors / patient party Equipment maintained in CCU 1) Monitor 2) Ventilator 3) Defibrillator 4) ECG and other equipments used in CCU Equipment maintained in specialized CCU 1) Intra Aortic Balloon Pump 2) Photo Therapy Machine Maintenance of uninterrupted gas and power supply system and their utilization 1)

Continuous wall flow oxygen 2) Compressed air 3) Suction Apparatus (Vacuum Pump) 4) Uninterrupted Power Supply (UPS) line.

SEMESTER III

HAM-202-Human Resource Management And Drug Management

Objective:

This course intends to develop understanding of the students of concepts, techniques, theories and practices of human resource management in hospital and about the general information of available drugs related issues applicable in hospital industries.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.

Unit - I

HRM 1 - To define different rules and regulations which are necessary in a hospital and personnel management. Rules & Regulation as applicable to Hospital employees, Hospital need assessment: Creation of new post & abolition of existing post, Job description and full utilization of manpower. Cost effectiveness

UNIT-II

HRM 2 Selection & Recruitment procedure, Wage & Salary. Qualification, Experience, Pay fixation, E.B., Manpower Need Assessment and Performance Indicator Tools, Manpower Audit in Hospital Care Industry, Training Need Assessment & in-service training, Code of Conduct and disciplinary proceeding

Unit - III

Drug management - To understand different aspects of drug management and quality control, Drug Management; Present scenario, Hospital Pharmacy Licenses, Drug Licenses, Narcotics drugs, Purchase of drugs and other consumable materials.

UNIT-IV

Drug Storage, Pharmacy Billing Computerized drug management system, Rational use of drugs and Prescription Audit, Spurious drugs, Banned drugs, Procedure of Drug Indenting, On time Drug dispensing, Inventory Control – ABC, VED, SDE, FSN Analysis ,Methods of ordering –Two bin system (Lead Time, Buffer stock, Reorder Level), Cyclic System

References:

- Materials Management - Gopalakrishnan
- Matgerials Management - A. C. Dutta
- CDA Rules - Navi Publications
- Law of Employment and Services in India - Kamal Law Book House
- Fundamental Rules-Navi Publications
- Supplementary Rules- Navi publications

HAM-203-Nursing Administration and Hospital Management

Objective:

To familiarizes the role of nursing and its administration in the existing hospitals & its growth.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.

Unit - I

Nursing Administration - To study the role of nursing administration in health care delivery, Introduction to Noble Nursing Profession, Nursing organization structure, Nurses – Doctors and Nurses – Patients' relationship

UNIT-II

Nurses: A dedicated social and professional entity, Staffing norms in various types of hospitals and departments, Recent trends in nursing profession and education, Specialization in nursing practices

Unit - III

Hospital Management - To visualize into the future the needs and expectation of the community from the hospitals, Treat your patients and treat also like your guest, Changing mind set of patients

UNIT-IV

Hospitality Management, Aims and objectives of Hospitality Management (Commercial point), Methods of Hospitality Management in a Hospital set-up, Attractive look, Effective conversation, Multi lingual, smart dress. Role of Hospitality Management in a Hospital set-up

References:

- Srinivasan A.V. (ed), Managing a modern hospital, Response Books, New Delhi.
- Anand K.K., Hospital management, Vikas Publishing, New Delhi,
- Arun Kumar, (ed) Encylopedia of Hospital Administration and Development, Anmol Publications, New Delhi.
- Park K, Textbook on Hygiene and Preventive Medicine.

HAM-204-PATIENT CARE AND BEHAVIOUR

Objective:

The Objective is to make the learner to understand the psycho-social aspects of the patient and resulting behaviour in a stressful condition.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.

UNIT-I

Introduction: Patient Rights – Patient Behaviour – Models of Patient Behaviour – Patient Motivation – Patient Perception – Attitudes – Attitude Change – Personality, Patient Involvement and Decision Making, Reference Group Influence – Opinion Leadership – Family Decision Making. Policies and Procedures of the Hospitals for patients and personnel: Service Buying Behaviour – Psychographics – Lifestyles – Information Search Process – Evaluating Criteria Audit of Patient Behaviour.

UNIT-II

Patient Care: Introduction, Importance of improving the quality care of patients, role of natural and human resources in patient care management, patient counseling: for surgical procedures, for treatment, grief counseling; protocols, Medicare standards.

UNIT-III

Hospital Administration: Role of Medical Superintendent, Hospital Administrator, Resident Medical Officer, Night Duty Executive; Public and guest relation; importance in patient care, information regarding patients, code of press relations, medical information, patient information booklets, attendants' management.

UNIT-IV

Legal Responsibilities: Essential documents, state licensure, civil rights, authority of examination, treatments, autopsy, responsibilities of medical staff, tort liability, insurance, use of investigational drugs. General policies and procedures of the hospitals for patients and personnel. Need, legal implications, Pollution Control Board Act, Safe Collection, segregation, disposal, dumping, incineration and training.

References:

- Liewellyne Davis and H.M. Macacaulay, Hospital Administraton and Planning, JP Brothers, New Delhi.

- S.G. Kabra, Medical Audit
- Arun Kumar (ed) Encyclopedia of Hospital Administration and Development, Anmol Publications, New Delhi.
- Srinivasan, A.V. (ed), Managing a Modern Hospitals, Response Books, New Delhi.
- Environment Management Systems, ISO 14000 Documents.
- Syed Amin Tabish, Hospital and Health Services Administration Principles and Practice, Oxford Publishers, New Delhi.
- Liewellyne Davis and H.M. MacCaulay, Hospital Administration and Planning, JP Brothers, New Delhi.

HAM-205-MANAGEMENT OF INFORMATION SYSTEM IN HOSPITALS

Objective:

To make the learner to understand MIS as a managerial decision making tool and to know the sources and compiling of MIS.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.

UNIT-I

Introduction to Management Information Systems: Decision Making Process Techniques – Major Trends in Technology in Decision Making – Computerized data processing – Decision Support Systems – Expert System – Executive Information System – Health Management Information System.

UNIT-II

Health Records: The World of Informatics The Future of healthcare technology - Functions of the health record – Changing functions of the patients record – privacy and confidentiality and Law – Advantages of the paper record – Disadvantages of the paper record – Optically scanned records – The Electronic health record – Automating the paper record – Advantages of the HER – Disadvantages of the HER – Bedside or point-or-care systems – Human factors and the HER – Roadblocks and challenges to HER implementation.

UNIT-III

Telemedicine: Telehealth – Historical perspectives – Types of Technology – Clinical initiatives – Administrative initiatives – Advantages and Barriers of telehealth – Future trends – Summary – The Future of informatics; Globalization of Information. Technology – Electronic communication – Knowledge management – Genomics – Advances in public health – Speech recognition – Wireless computing – Security – Telehealth – Informatics Education – Barriers to Information Technology implementation.

UNIT-IV

Software Applications in Health Care: Awareness on the application of computer software packages in various functions of hospital – Internet and Intranet and their application in healthcare. Practicals of Software Applications: One way ANOVA using Statistical Software.

References:

- Green. E. Paul. Danald S. Tull, Gerald Albaum, Research for Marketing Decisions, Prentice Hall, New Delhi.
- Ghosal, A., Elements of Operations Research, Hindustan Publishing Corporation, New Delhi.

Paper - HAM 206

Hospital Training and Viva Voce

At the end of the semester, students should compulsorily undergo two week Hospital training (HAM 206) and the relevant certificate and project report has to be submitted.

Practical training includes the following:

Job Roles:-

Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.

Pharmacy Services Involvement in:- 1) Space requirement as per act 2) Lay out 3) Storage facility – different set up for different type of drugs 4) Equipments 5) Organizational structure (role and function) a) In-charge b) Pharmacist and Chemist c) Sales Technician d) Cash Counter e) Computer Operator / Office Assistant f) Attendant 6) Purchase of drugs – disposables, consumables etc. 7) Purchase of food items 8) Quality control 9) Proper storage of drugs 10) Sales policy 11) Record maintenance 12) Preparation of Accounts 13) Pharmacy Audit

Stores - Observation and participation in:- Different stores like Medical Stores, Linen Stores, Housekeeping Store, Maintenance Store, Printing & Stationary Store etc. 1) Space requirement

2) Lay out (as per category of store) 3) Storage facility, proper set up for particular type of medical and non-medical items 4) Issuance policy 5) Safety measure 6) Practical application of E O Q 7) Minimum stock level 8) Deployment of staff like; a) In-charge b) Store keeper c) Record maintenance clerk d) Attendant / Delivery person e) Security etc. 9) Proper storage procedures 10) Indents 11) Issue of item as per indent 12) Record keeping of issued materials as well as balance stock 13) Delivery system 14) Physical verification of stored items periodically Laundry & Linen Services Observation of 1) Location 2) Work flow 3) Physical facilities of Laundry a) Receiving, Storing, Sorting, washing area b) Central Disinfection Area c) Cleaned Linen processing room d) Laundry Manager's Office e) Staff Room f) Sewing and inspection area g) Supply storage room h) Solution Preparation area i) Cleaned Linen issue area j) Natural ventilation and light k) Free Linen movement area 4) Other facilities a) Adequate Water supply b) Drainage system c) Adjacent power supply d) Steam e) Compressed air 5) Arrangement of equipments 6) Soiled linen receiving from ward, OT, other areas 7) Process of sorting and cleaning, washing, drying, ironing 8) Collection and storing process of cleaned linen 9) Sorting of discarded linen and sending them to the Store 10) Process of linen distribution 11) Maintenance of hygiene 12) Infection control process 13) Process of staff training 14) Process of co-ordination among the related departments 15) Manpower for Laundry (a) Laundry Manager (b) Supervisor (c) Supporting staff

Kitchen & Dietary Services - Overview of kitchen lay out and its adjacent areas, Physical facilities, Receiving area, Washing area, Preparation area, Cooking area, Storing area, Food Distribution area, Garbage disposal area, Office of Kitchen Manager, Office Kitchen Staff

Semester IV

HAM-208-Public Relations

Objective:

This course intends to develop and sharpen the skills of effective public relation for better positioning of the hospital and communication of healthcare services and facilities to the public.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.

UNIT-I

Public Relations 1 - Introduction to Public Relations, Meaning of Public Relations, Classification of Public from Healthcare Marketing Perspective, Evaluation of Public Relations, Public Relations Process, Identifying the Relevant Publics, Measuring Images and Attitude of the Relevant Public, Establishing Image and Attitude Goals for the Key Publics

UNIT-II

Public Relation 2 - Developing Cost Effective Public Relations Strategies, Implementing Actions and Evaluating Results, Community Opinion Surveys to Assess the Image of an Organization

Unit - III

Public Relation 3 - A Model Questionnaire used in Healthcare Services, Methodology of the Study, Public Relations Tools, Materials: Written and Audiovisual, Media: News, Events, Speeches and Telephone Information Services

Unit - IV

Public Relation 4 - Health Service Public Relations Officer, Profile of Public Relations Officer, Changing Role and Responsibilities of Health Service PRO.

References:

- Philip Kotler and Roberta N. Clarke, Marketing for Healthcare Organizaions (Prentice Hall Publication)
- Roger Silver, Health Service Public Relations (Radcliffe Medical Press Ltd., Oxford, 1995) ISBN 1-85775-028-4
- John F. O□Malley, Healthcare Marketing Sales and Services: An Executive Companion (Health Administration Press) ISBN 1-56793-150-2
- G.D. Kunder, How to Market Your Hospital Without Selling Your Philosophy (Prism Books Pvt. Ltd., Bangalore)

HAM-209-MANAGEMENT OF OUT-PATIENT DEPARTMENT (OPD)

Objective:

The objective of the course is to equip the learner with the different organizational steps involved in management of the Out-Patient Department.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.

UNIT-I

Evolution of Out Patient Department – Planning of Out Patient Department – Objectives – Functions of Out Patient Department – Organization Structures

Unit - II

Day Care Centres – Community Health Care Centres – Physical Medicine & Rehabilitation Centre – Performance Evalution of Out Patient Departments – Conclusion – Unit Review Questions – Reference Books.

Unit - III

Emergency Services – Managerial Issues in Emergency Department – Staffing of Emergency Department – Checklist Management. Patient Care in Out Patient Department – Emergency Operation Theatre Management –Planning & Organization of Minor Operation Theatre.

UNIT-IV

Departments of Out-patient Departments – Planning of Departments. Diagnostic Services of Out Patient Department – Secretarial Services in Management of Out Patient Departments – Referrals & Cross Referrals – Treatment.

References:

- S.L. Goel, Healthcare Management and Administration, Deep & Deep Publications Pvt. Ltd. New Delhi.
- Srinivasan, A.V. (ed.), Managing a Modern Hospital, Chapter 12, Response Books, New Delhi.

HAM-210-Healthcare Marketing

Objectives:

The aim of this course is to enable the participants to understand the concept of marketing and public relations, and applying them in the Healthcare organizations. At the end of the course the participants would be able to Accept marketing as a managerial process Apply marketing function in health care organizations without conflicting the professional ethics of the clinical professions and ultimately aiming as customer satisfaction Define the Market, targeting, focus everything from customer point of view and be customer oriented through consumer analysis and awareness creation Do marketing research, which would enable effective service planning with the information about the market, and community health needs Manage demand and supply; Analyze wholeness of customer costs and price sensitivity; develop access to services; and Use promotional tools as communication tools which help to create awareness, knowledge and conviction without commission or incentive Use Social Marketing in marketing ideas and causes, which would change the behavior of target group Practice Health Services Public Relations.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.

Unit I

Application of Marketing in Healthcare - Marketing Concept of Marketing, Importance of Marketing to Healthcare Organizations, Challenges in Practicing Marketing in Healthcare Industry, Marketing Intelligence, Information and Research System, Marketing Organization: Chart, Fulltime Staff and Hiring Consultants.

Unit II

Market (Individual and Organizational) Analysis - Market Measurement and Forecasting
Defining Market, Measuring Current Market Demand, Forecasting Demand, Market Segmentation and Targeting, Steps in Segmentation and Targeting, Target Marketing
Consumer Analysis, Consumer Buying Process, Organization Buyer Behaviour, Consumer Adoption Process

Unit - III

Marketing Mix Decisions - Product (Service) Decisions, Nature and Characteristics of Services
Marketing Approaches to New Services Development, Service Mix Decision, Service Item Decision, Service Life Cycle Decision, Demand Generation, Pricing Decisions, Pricing Objectives in Healthcare, Pricing Strategy, Present and Future Situation, Place Decisions, Major Distribution Decisions, Strengthening Referral System, Promotion Decisions, Sales Force in Healthcare Organizations, Advertising in Healthcare Industry, Sales Promotion Practices in Healthcare Organizations, Publicity Practices

Unit - IV

Marketing Strategies, Service Portfolio Strategy, Market Expansion Strategy, Target Market Strategy, Price Quality Strategy, Competitive Positioning Strategy, Marketing Mix Strategy, Social Marketing, Steps in Social Marketing Cognitive, Action, Behaviour and Value Changes

References:

- Philip Kotler and Roberta N. Clarke, Marketing for Healthcare Organizaions (Prentice Hall Publication)
- Roger Silver, Health Service Public Relations (Radcliffe Medical Press Ltd., Oxford) ISBN 1-85775-028-4
- John F. O□Malley, Healthcare Marketing Sales and Services: An Executive Companion (Health Administration Press) ISBN 1-56793-150-2
- G.D. Kunder, How to Market Your Hospital Without Selling Your Philosophy (Prism Books Pvt. Ltd., Bangalore)

HAM-211-Medical Equipment and Floor Management

Objective:

To make aware the equipment management process to the students about various biomedical equipments used in healthcare industries & its solution.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.

Unit - I

Equipment management process and its various components and their roles in hospital system - Establish equipment management procedure for a hospital, List of common Medical Equipments used in Hospital, Justification of purchase proposal, Hospital Need Assessment

Unit - II

Equipment selection guideline, Estimation of cost and Q.C. Planning, Purchase / Installati8on / Commissioning of Medical Equipments, Replacement of old equipments and Buyback Policy, Estimation of Breakeven point and Profit – Projection in hospital budget, Medical Equipment Maintenance (In-house and AMC), Local, National and International availability of Medical Equipments

Unit - III

Management of wards - Different types of ward , Position of Nursing Station, Ward facilities, duties and responsibilities of ward staff, Manpower Need Assessment in Hospital Ward

Unit - IV

Management of OT - General concept on OT design and function of OT, Advantages and importance of zoning of OT, Advantages of Centralized OT complex

References:

- Materials Management - Gopalakrishnan
- NIHFW Monograph - Govt. of India
- Hospital Management - Tabish
- Hospital Management - Goel
- Hospital Management McGibony
- Paramedics-Six in One, Jaypee Brothers

Paper - HAM 212

Hospital Training and Viva Voce

At the end of the semester, students should compulsorily undergo two week Hospital training (HAM 212) and the relevant certificate and project report has to be submitted.

Job Roles:-

Purchase Assistant, Nursing Administrator, Patient Care Coordinator, Data Analogist, Marketing Assistant, Equipments Technician, Floor Manager.

Practical training includes the following:

Pharmacy Services Involvement in:-

OPD Services Involvement in :- Layout of Reception Desk in OPD, Registration and department wise OPD Card segregation, Location of the concerned Department, Reception of patients, Physical facilities in OPDs, Close supervision of Doctor's Chamber for the followings:-

Availability of - Prescription Pad, Stethoscope, View Box, Bed Trolley, Weigh Machine, Torch Light, Gloves, Liquid soap and towel; Records maintenance of OPD, Supervision of patient waiting area and its seating arrangement, Adequate communication with other departments / units / wards etc., Public Addressing System, Deployment of Staff like Jr. Doctor, Para Medical Staff, Receptionist, Assistant etc.

Public Relations Office

(a) General - Helping patients regarding the followings: 1) Information – Over phone / e-mail / letter / fax / across the table 2) Dispatch reports for outstation patients 3) Providing different types of brochure 4) Liaison between doctor and patient 5) Appointment for doctors 6) Estimation of treatment expenditure 7) Billing status of IPD patient 8) Visiting indoor patients to enquire about their facilities 9) Availability of Public Relations Officer everyday including Sundays and holidays (b) Corporate 1) Coordination of admission of company patients and patient having medical insurance for cashless treatment. 2) Ensuring validity of ID card and referral letter 3) Filling up the pre-authorization letter and facsimile the patient case history, investigation report, clinical notes etc. to the concerned TPA for verification and approval. 4) Provide different additional queries regarding the health status and treatment procedure of the patients to concerned TPA. 5) Informing the admission counter about the status of the patient awaiting admission. 6) Close liaison with concerned department or doctor of Corporate Houses 7) Coordination with Billing and other Depts. for collection of paper documentation

Marketing Observation / involvement in:- 1) Office Layout 2) Different types of advertisement for projection 3) Fixture required for marketing (a) Banner (b) Leaflet (c) Hoarding (d) CD / VCD / DVD 4) Media Management (Print, Electronic) 5) Event Management (Camp, Seminar, CME) 6) Institutional / Corporate marketing 7) Direct marketing 8) Channel / Franchise marketing

9) Data analysis 10) Patient feedback 11) Organization a) Head of Marketing b) Departmental Manager c) Regional and Territory Manager d) Marketing Executive e) Office Assistant

Wards Observation of:- 1) Ward Composition - Beds Patient Ratio 1) Layout 3) Types of ward 4) Physical facilities a) Nursing Station b) Doctor's room c) Dirty utility room d) Dress Changing room e) Sanitary facilities f) Dressing room for minor treatment g) Isolation room h) Ward Pantry i) Day room j) Barrier Nursing k) Other facilities i) Ward side Laboratory ii) Linen Store iii) Drug Store etc. 5) Other Equipment facilities a) Electronic Monitoring Devices b) Ventilation Devices – Natural & Mechanical c) Ward Lighting d) Calling Bell e) Fire Alarm System 6) Organization a) Resident Medical Officer b) Nursing Superintendent c) Floor Co-coordinator d) Nursing In-charge e) Nursing Personnel f) Nursing Trainee g) Ward Boy 7) Communication facilities 8) Methods of patients' record keeping in wards / technical / medical details 9) Admission and Discharge procedures 10) Billing system / generation of bills based on bed head ticket entry

11) Cleanliness 12) Duty arrangement of various medical and para medical staff

Semester V

HAM-302-MEDICAL ETHICS AND LEGAL ISSUES IN HOSPITAL

Objective:

To familiarize the students about legal issues involved in hospital management.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.

UNIT I

MEDICAL ETHICS - Introduction to Legal aspects in health care: health as a fundamental right; Hospital services and law: Medical ethics including Hippocratic Oath, declaration of Geneva and international code of medical ethics and violation of these ethics; informed consent

Unit II

RIGHTS AND RESPONSIBILITIES OF PATIENTS - Rights and responsibilities of patients: confidentiality issues with respect to contractual obligations towards the patients, electronic medical records and Health. Liability and accountability towards patients and third parties and employees (vicarious liability)

UNIT III

MEDICAL NEGLIGENCE - Medical negligence -definition, The Bolam test, Civil and criminal negligence, relevant sections of Indian Penal code, defense against medical negligence; The Consumer Protection Act- Ethics Introduction – its scope, various levels of consumer courts (District forum, State Commission, National Consumer Dispute Redressal Forum) and procedure followed in each; Professional Indemnity; Legal Implications in Medical Emergencies; Euthanasia- its types and the international & national Perspective; the concept of Living Will- Medical ethics.

UNIT IV

HEALTH CARE LAWS - A brief account of various laws related to healthcare: -Human Organs Transplantation Act, Inquest and its proceedings, The Drugs and Magic Remedies Act, Registration of Birth and Death Act, PNDT act (1994) and MTP Act (1971). Regulatory framework of hospitals – The Indian Medical Council act; Regulatory requirements to set up a Blood Bank; Legal aspects of Assisted Reproductive techniques

RISK CONSENT - Preventive Steps for Doctors/Hospitals to A void Litigation -Consent Form - Life support - High Risk Consent - Post Mortem

Reference:

- Malik, Vijay. 'Law Relating to Drugs and Cosmetics', Eastern Book Company.
- Pozgar, George. 'Legal aspects of health care administration', Jones & Bartlett Publishers.
- Green, Michelle and Mary Jo Bowie. 'Essentials of health information management: Principles and practices', Cengage Learning.

HAM-303-HOSPITAL WASTE MANAGEMENT

Objective:

The Objective of the Course is to familiarize the learner with the importance, techniques and the procedures involved in the management of Hospital Waste.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.

UNIT-I

Hospital Hazards: Meaning – Types – Physical–Biological Mechanical – Psychological – Its Impact on Employees – Preventive measures. Hospital Hazards Management: Meaning – Need – Principles – Purpose.

UNIT II

INDECTION Control of Hospital Acquired Infection: Types of Infection – Common Nosocomial Infection and their Causative Agents – Prevention of Hospital Acquired Infection – Role of Central Sterile Supply Department – Infection Control Committee – Monitoring and Control or Cross-Infection – Staff Health.

UNIT-III

Biomedical Waste Management: Meaning – Categories of Biomedical wastes – Disposal of biomedical waste products – Incineration and its importance – Standards for Waste Autoclaving, Micro Waving and Deep Burial – Segregation – Packaging – Transportation – Storage.

UNIT IV

Human Waste Disposal and Sewage Disposal: Diseases carried from excreta – Sanitation barrier – Methods of Excreta disposal – Sewage wastes: Meaning – Composition – Aims of Sewage disposal – Decomposition of Organic Matter – Modern Sewage Treatment – Drawbacks of improper disposal of wastes – Solid and liquid.

References:

- Sharma – Holistic approach to Hospital Waste Management published by Dept. of Hospital Administration – AIIMS, New Delhi.

HAM-304-HOSPITAL SAFETY SYSTEM AND HEALTH CARE STANDARDS

Objective:

To familiarize the students about the hospital safety and health care systems.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.

UNIT I

HEALTH ADMINISTRATION - Health administration in India- health care delivery system- national health policy.

NATIONAL HEALTH ISSUES - National Health Programmes- Tuberculosiss control Programme, DOTS- Programme for control of Blindness- Family welfare progrmme- AIDS control programme, role & functions of National AIDS Control Organisation (NACO).

UNIT II

HEALTHCARE REPORTS - Review of reports on Healthcare- Epidemiological Triad, Levels of Disease Prevention- Disaster Management/Disaster Plan -Fire fighting, Dealing with crisis situations, Natural disasters -floods, earthquakes etc., Mob violence against medical establishments, Bomb threat, Terrorist strike, Political agitation, Mass casualties

UNIT III

HOSPITAL ENGINEERING SERVICES - Engineering Services: Maintenance of Building, Campus & Utilities- Biomedical services, Electrical safety, chemical safety, Fire safety. Quality Management in health care Service and Process quality Cost of quality, Quality control -Statistical methods in hospitals for control of service quality-Standard Operating Procedures (SOPs). -ISO, ISO standards and their implementation- Accreditation of hospitals -ISO, NABH, JCI

UNIT IV

HEALTH STANDARDS - Health standards Health Level (HL), DICOM. Insurance companies and TPAs: Insurance policies and cover - Personal accident insurance benefit- Claims procedures- Administration of patient related schemes- Medical insurance -cashless benefit, reimbursement- CGHS, ECHS, CSMA, ESIC

References:

- K .Park, 'Park's text book of Preventive and Social Medicine', 16th. Ed. Jabalpur, India: M/s Banarasidas Bhanot, 259-67, 2002.
- Madhuri Sharma, 'Essential of Hospital Support Services & Physical Infrastructure', Jaypee Brothers, New Delhi.
- G.P. Mogli,'Medical Records Organization& Management', Jaypee Brothers, New Delhi.
- Branson, Joan Cameron, and Margaret Lennox. 'Hotel, hostel and hospital housekeeping', Edward Arnold.

HAM-305-Hospital Material Management

Objectives:

The purpose of this course is to learn the specific methods of materials planning, purchasing, processing, stocking and dispensing including equipment purchasing and maintenance.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.

Unit I

Materials Management - Introduction, Definition and Function, Goals and Objectives of Materials Management, Materials Cycle, Functions of Materials Manager, Problems and Issues in Hospitals, Information Systems for Materials Management

UNIT II

Purchasing - Objectives and Elements of Purchasing, Purchasing System, Purchasing Cycle, Purchase Procedures, Legal and Ethical Aspects, Conditions of Contract, Financial Rules, Arbitration
Equipment Purchase and Maintenance - Planning and Selection of Equipment, Import of Equipment, Equipment Utilization and Operation, Equipment Repair and Maintenance, Equipment Audit

Unit III

Inspection, Storage and Distribution of Materials - Planning Consideration of Stores, Inspection and Verification of Materials, Storage of Materials, Distribution of Materials, Condemnation and Disposal.

Unit IV

Scientific Inventory Management - Codification and Standardization, Value Analysis, Inventory Control, Lead Time, Safety Stock and Reorder Level, Economic Order Quantity (EOQ)
Selective Controls, Case Studies on Inventory Control

References:

- Shaki Gupta and Sunil Kant, Hospital Stores Management: An Integrated Approach (Jaypee Publications, New Delhi, India)
- WHO, Maintenance and Repair of Laboratory, Diagnostic, Imaging and Hospital Equipment (WHO, Geneva)
- Murriel Skeet and David Fear, Care and Safe Use of Hospital Equipment (VSO, UK)

- P. Gopalakrishnan and M. Sundaresan, Materials Management : An Integrated Approach (Prentice – Hall of India Pvt. Ltd., New Delhi) ISBN 81-203-0027-0

Paper - HAM 306

Hospital Training and Viva Voce

At the end of the semester, students should compulsorily undergo two week Hospital training (HAM 212) and the relevant certificate and project report has to be submitted.

Practical training includes the following:

Job Roles:-

Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.

Security Services Observation of:- 1) Security room and posts 2) Security services in the hospital 3) Role of Security during patient visiting hours 4) Traffic control inside the hospital 5) Maintenance of fire precautions 6) Process of medico legal activities Central Sterile Supply Department (CSSD) Observation of 1) Location 2) Name and function of equipments 3) Items processed by Central Sterile Supply Department 4) Sterilization technique a) Heat b) Gas c) Liquid d) Ionizing radiation e) Others 5) Work flow 6) Clean zone, dirty zone 7) Pooling of materials, equipments 8) Function of different equipments used for sterilization 9) Process of receiving and distribution of materials 10) Technique of Gauge folding 11) Cleaning, drying and packing of sterilized materials 12) Segregation of Dirty zone and clean zone 13) Process of receiving and distribution of materials 14) Quality control checking of sterilization in collaboration with Micro-biology Lab. 15) Organization a) CSSD Supervisor b) Technologist c) Technical Assistant d) Para medical staff e) Attendant 16) CSSD Committee

Operation Theatre Observation / involvement in:- 1) Location 2) Zoning of Operation Theatre 3) Infrastructural facilities 4) Centralized and decentralized Operation Theatres 5) Equipment requirement 6) Procurement and maintenance including annual maintenance contract 7) Functions and policies of Operation Theatres 8) Manpower requirement a) Medical Superintendent b) In-charge c) Anaesthetist c) Nursing personnel d) Technician e) Ward Boy 9) Duties and responsibilities including standard operation procedures 10) Safety procedures 11) Methods of checking operating rooms for readiness to receive patient 12) Periodical sterilization / fumigation 13) Sterile supply 14) Equipments procurement and maintenance

Telemedicine 1. Infrastructure – both end – (1) Hospital end (2) Nodal Centre 2. Requirement

a) Computer Monitors b) Cameras c) Internet connection 3. Observation of telemedicine techniques and taking part in the process 4. Manpower requirement a) Medical Officer b) Technician

Semester VI

HAM-308-GLOBAL HEALTHCARE SCENARIO & INSURANCE

Objective:

The Student is expected to understand the nuances of Insurance and in particular the Health Insurance.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.

UNIT-I

International Health Scenario - International health organization and their role. International Health Regulations, & International health insurance schemes, concept of HMO, PPO, TPA etc. International Aid agencies in healthcare. Medical/Health Insurance, National Community Health Insurance Schemes, Roles of Governments, NGO's, Private Sectors in healthcare financing and distribution, national health insurance schemes

Unit II

Health Tourism concept and history of health tourism and medical health tourism, Process, Legal and Ethical issues, Risks, Common fields of care in Medical Tourism, Merging with pleasure Tourism, Issues of Safety and Security, Concerns, Prevention of Malpractices.

UNIT-III

Insurance - Introduction – Economics of Life and Health Insurance – Importance, Socio-political realities –Insurance Terminology, Health Policy vis-à-vis Health Insurance Policies – Indian Scenario – Different Products –Demand and Scope – Limitation, Administration of Health Insurance Schemes like CGHS & ESI and Social Security Measures.

Unit IV

TPAs, Governing Mechanisms including IRDA, Health Insurance Taxation – Four General Insurance Companies - Standardization and grading of Hospital Services – Role of Vigilance and Real-time

information about the services, Health Insurance Providers – Government and Private – Micro-insurance, The role and responsibilities of provider – insurer-patient and the regulatory agencies.

References:

- Gupta P.K, Insurance and Risk Management, Himalaya Publishing House.
- My Collection of HI – Material and Books.
- Insurance Exam – Material
- American Books – Managed Care On History of Health Insurance – Origin –Development & Growth.
- Insurance – Compendium – Given by Venugopal.

HAM-309-HOSPITAL PLANNING AND MANAGEMENT

Objective:

To provide the knowledge of planning, designing and safety management in hospital services.

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.

UNIT I

PLANNING AND ORGANIZATION OF THE HOSPITALS - Roles of hospital in healthcare-hospital planning and design-outpatient Services the nursing unit-intensive care unit-nursing services-effective hospital management directing and leading-controlling – financial management.

Unit II

CLINICAL AND ADMINISTRATIVE SERVICES - Radiology and imaging services-laboratory services-operation theatre suite pharmacy central sterile supply department- hospital infection-materials management-evaluation of hospital services.

UNIT III

DESIGNING OF HOSPITAL SERVICES - Engineering department - maintenance management-clinical engineering electrical system- air conditioning system- water supply and sanitary system centralized medical gas system-communication system- solid waste management and transportation.

HOSPITAL SUPPORT SERVICES AND SAFETY SYSTEMS - Designing support services and safety management -Admitting department- medical records department- food service department laundry and linen service-housekeeping volunteer department- safety in hospital fire safety- Alarm system-disaster management.

Unit IV

INFECTION CONTROL - Importance of infection control-hand hygiene-aseptic techniques-isolation precautions-disinfection and sterilization-clinical laboratory standards to infection control health care workers safety.

References:

- G D Kunders, 'Biomechanics: Hospitals, facilities planning andmanagement*',* Tata Mcgraw Hill.
- B M Sakharkar, 'Principles of hospital administration and planning', Jaypee Brothers Medical Publishers Pvt Limited, 2nd edition.
- Sanjiv Singh, Sakthikumar Gupta, Sunil Kant, 'Hospital infection control guidelines, principles and practice', Jaypee Brothers Medical Publishers Pvt Limited, First edition.

HAM-310-COMPUTERS IN MEDICINE

Objective:

To familiarize the student with use of computers in medicine

Instructions:

- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.

UNIT I

INTRODUCTION - Computer hardware and software - programming languages - use in medical field - need of computing hospitals - cost effectiveness - help of computerization to physicians.

Unit II

PATIENT DATA BASE MANAGEMENT - Computerized medical records - security. Computer in clinical laboratory –database approach - automated clinical laboratory and analysis - computerized specimen analysis- analysis of ECG, EEG and EMG. Chromosome analysis by computer - computerized cytology and histogram - automated scanning for cervical cancer.

UNIT III

MEDICAL IMAGING - Basics of computer assisted medical imaging: Nuclear medicine - digital subtraction radiography – computerized ultra sonography – X-ray, CT, Nuclear magnetic resonance. Basics of computer assisted medical decision making - general model algorithms - fuzzy set theory - cognitive set theory - cognitive models - QMR, KES and TIA.

Unit IV

COMPUTER IN INTENSIVE CARE UNITS - Metabolic balance up keeping- pulmonary function evaluation- Cardio vascular evaluation- Computer assisted therapy- computer for case of renal disorders.

COMPUTER AIDS - Computer aids for the handicapped: Basic discussion with examples- introduction to computer assisted instruction in medicine- ISDN in medicine.

References:

- R.D. Lele, 'Computers in medicine', Tata McGraw-Hill Publishing Company Limited, 2nd reprint.

HAM-311-RESEARCH METHODOLOGY

Objectives:
At the end of this course, the students will be able to Design research methods, collect and analyze data, Prepare research report and independently conduct research in their field of study / area of work.

Instructions:
- The syllabus of this paper has been divided into FOUR units.
- Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
- All questions carry equal marks.

Job Roles:-

Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.

Unit I

Introduction, Research Planning and Design Meaning, Purpose, Types and Methods of Research Research Process Steps in Research Planning Problems: Components, Selecting and Defining Problem Purpose, Objectives and Hypothesis Research Strategies Ethical Aspects in Health Research Economic Considerations Data Collection Plans: Questioning, Information Gathering and Sources Plans for Analysis of Data Proposal Writing Research Design in case of different researches

Unit II

Sampling Methods Census Vs Sampling Universe, Sampling Frame, Sampling Design, Errors, Precision and Confidence Level Principles of Sampling Steps in Sampling Design Systematic Bias & Causes of Bias Types of Sampling Design Merits and Limitations of Sampling 560
Data Collection Primary and Secondary Data Methods of Collecting Primary Data : Questioning and Observation Prerequisites / Basic Tenets of Interviewing Designing a Questionnarie Pre-testing / Pilot Study

Unit III

Data Analysis : Biostatistics Functions, Scope and Limitations Presentation of Data Measures of Central Tendency Measures of Variation Correlation and Regression Analysis Forecasting and Time Series Analysis Probability Sampling Estimating of Parameters Tests of Hypothesis Chi Square Test Statistical Quality Control

Unit IV

Interpretation and Report Writing Reasons, Techniques and Precautions in Interpretation Significance of Report Writing Steps in Report Writing Layout of a Research Report Types of Reports University Prescribed Guidelines Precautions

References:

- CR Kothari, Research Methodology Methods and Techniques (Wiley Eastern Ltd., New Delhi)
- PSS Sundar Rao and J Richard, An Introduction to Biostatistics – A Manual for students in Health Sciences (Prentice – Hall of India Pvt. Ltd., New Delhi)
- Stephen Polgar and Shane S Thomas, Introduction to Research in the Health Sciences (Churchill Livingstone Publications)
- BK Mahajan, Methods in Biostatistics (Jaypee Brothers Medical Publishers P. Ltd., New Delhi) SP Gupta and MP Gupta, Business Statistics (Sultan Chand & Sons, New Delhi)

HAM-312- HOSPITAL TRAINING

Objective:-

- Allow students to explore the prevalent commercial/industrial practices in order to reinforce theoretical knowledge with practical experience.

Job Roles:-

Hospital Planner, Hospital Consultant, Hospital Administrator, Health Service Manager, Legal Assistant, Materials Coordinator, Business Operational Specialist, Health Service Project, Administrator.

At the end of the semester, students should compulsory undergo two week Hospital training (HAM 312) and relevant certificate and report has to be submitted.

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